A Structural Break, Not a Weaker Cohort

Explaining Albania's 2022 PISA Collapse with
Survey-Weighted Machine Learning

PISA 2009–2022 Data Science Project

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Abstract

Albania's share of 15-year-olds below PISA mathematics Level 2 rose from 42% in 2018 to 75% in 2022, the sharpest regression in the assessment. We ask whether machine learning on background questionnaire data can identify at-risk students, *why* they are at risk, and whether the 2022 collapse is merely a lower-scoring cohort or a change in the risk structure itself. Using all five Albanian PISA cycles (2009–2022; 27,042 students) and eight comparison countries (323,121 students), and applying the survey methodology PISA requires (final student weights on every estimate, plausible values combined by Rubin's rules, and balanced repeated replication for design-based standard errors), we report four findings. First, a domain classifier separates 2018 from 2022 Albanian students with AUC 0.98 on background features alone: the collapse is a *structural covariate shift*, not just lower scores. Second, and this is our central methodological contribution, the accuracy ceiling that individual background variables hit (weighted cross-validated AUC 0.73, unmoved by hyperparameter tuning) is a *feature* ceiling, not a data ceiling: adding survey-weighted *school socioeconomic composition* lifts every model by $\sim$0.05 AUC to $\sim$0.78 and makes gradient boosters significantly outperform logistic regression. Third, the new $\sim$0.78 ceiling is then a genuine *data* ceiling: a multilevel model (school ICC ≈ 0.31) and a direct school-questionnaire linkage (no significant gain, all $p > 0.24$) converge on it, it recurs across all nine systems with its size tracking each system's between-school ICC ($r \approx 0.80$), and richer within-test process data lifts AUC to $\sim$0.86 only through a signal endogenous to the score, so the deployable ceiling holds. Fourth, the crisis is *broad-based*: Albania has the flattest SES gradient of the nine countries, its risk drivers (math anxiety, school composition) are shared with peers, and a difference-in-differences with a failed placebo shows the collapse is national rather than local to the 2019 earthquake. We release a calibrated, interpretable risk screener, framed as associational, not a policy lever; a naive threshold over-flags the poorest quintile (FPR gap 0.63), which group-specific thresholds close to 0.003 at a recall cost, and Manski bounds under 79% coverage leave the crisis far above 2018.

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1. Introduction

In four years, three in four Albanian fifteen-year-olds crossed the line into low mathematics proficiency. The OECD Programme for International Student Assessment (PISA) measures whether students can apply reading, mathematics, and science to real problems, and its *Level 2* boundary marks basic proficiency; students below it are the international "low performers" that education systems target. In PISA 2018, 42% of Albanian students fell below the mathematics Level 2 line. In PISA 2022, 75% did. No other participating system regressed so far in a single cycle. A drop of this magnitude invites an obvious reading: the 2022 cohort was simply weaker, and the country needs broad remediation. Whether that reading is right is not a semantic question. It determines whether policy should treat 2022 as more of the same, only worse, or as evidence that *who* is at risk, and *why*, has changed.

This paper treats the question as one of educational data mining, and it insists on the survey methodology that PISA requires. PISA is a stratified, clustered sample reported through final student weights, and each achievement score arrives as ten *plausible values*, multiple imputations of a latent proficiency. Analyses that ignore the weights do not estimate a population quantity, and analyses that collapse the plausible values to a single number understate their uncertainty. A large body of secondary PISA work nonetheless does both. Every estimate below is survey-weighted; plausible values are combined by Rubin's rules; and where the data permit (the 2015, 2018, and 2022 cycles carry 80 balanced-repeated-replication weights) standard errors are design-based.

From the student background questionnaire, socioeconomic status, home possessions, parental education and occupation, grade repetition, immigration status, school belonging, teacher support, math anxiety, and access to information technology, we ask three questions. Can a classifier identify Albanian students at risk of low mathematics proficiency, and how well? Which factors drive that risk, and do they differ from the drivers in comparable countries? And does the 2022 collapse represent a change in the risk *structure* that can explain the 2018-to-2022 reversal? The economic and structural conditions of Albania in this exact window, a destructive 2019 earthquake, an extended pandemic school closure, a thin remote-learning infrastructure, and sustained emigration of working-age adults and teachers, give these questions a concrete backdrop that we return to in Section 6.

Thesis. Albania's 2022 mathematics collapse is not a uniformly weaker cohort but a *structural break in the risk-generating process*, driven by a disruption to the supply of schooling rather than to household resources; the students at risk are identifiable from background data only up to a genuine ceiling of about 0.78 AUC that three independent methods confirm, their risk is broad-based rather than concentrated in the poor, and its drivers are the same ones that operate across the region.

Roadmap. Section 3 describes the five-cycle data and the design-based estimation. Section 4 sets out the leakage-safe, survey-weighted modelling and the covariate-shift diagnostic. Section 5 reports the four empirical findings: the structural shift (AUC 0.98), the feature-versus-data ceiling, the shared and broad-based driver structure, and the calibrated, fairness-audited screener. Section 6 then reads those findings against Albania's economic and structural conditions, arguing that a supply-side shock, not a household-income shock, best fits the evidence. Section 7 discusses implications and limits, and Section 8 returns to the collapse with which we began.

Contributions. Our central contribution is methodological and, we believe, transferable beyond this dataset: we show that the accuracy ceiling recurring across PISA prediction studies is not one phenomenon but two, a movable *feature* ceiling and a fixed *data* ceiling, and that the two can be told apart with convergent evidence rather than with another round of tuning. Applied to Albania, this lets us (i) show the 2022 collapse is a structural covariate shift rather than a level change; (ii) locate the background-data accuracy ceiling, demonstrate it is a feature ceiling by moving it with school socioeconomic composition, certify the new plateau as a genuine data ceiling through three independent routes, and show it both generalises across nine systems (its size tracking each system's between-school inequality) and withstands richer process data that only moves it through a signal endogenous to the test; (iii) characterise the crisis as broad-based and shared with regional peers, and test a single-region earthquake explanation with a difference-in-differences that instead identifies the collapse as national; and (iv) connect these statistical findings to Albania's economic and structural context, while delivering a calibrated, interpretable, coverage-bounded and fairness-mitigable risk screener that we argue should not be read causally.

2. Background and Related Work

PISA survey methodology. PISA reports each student's proficiency as ten plausible values drawn from a posterior latent-trait distribution, and every published statistic is weighted by the final student weight, with sampling variance estimated by balanced repeated replication (BRR) [14, 15]. Rubin's rules combine the point estimates and the between-imputation variance across the plausible values [19]. These are not optional refinements: using a single averaged score as a target understates uncertainty, and unweighted statistics do not estimate a population quantity. A large share of secondary PISA analysis nonetheless drops both conventions, which, as we argue below, is part of why the accuracy figures it reports should be read with care. We follow the manual throughout.

Machine learning on PISA background data. Predicting achievement or at-risk status from PISA background questionnaires is by now a well-populated literature, and it converges on a consistent picture in which tree ensembles dominate. Masci, Johnes and Agasisti [12] analyse PISA 2015 across nine countries with a two-stage tree-based model and recover socioeconomic status, grade repetition and school climate as leading correlates, while stressing that much of the variation sits at the school rather than the student level. On the most recent cycle, Djekourmane et al. [4] predict mathematics scores across six East Asian systems in PISA 2022 and find gradient boosting strongest: XGBoost reaches $R^2 \approx 0.57$ and a thresholded AUC near 0.81, with LightGBM close behind, and, tellingly, the linear baselines trail badly on R^2 yet come within a few points on AUC. Erdoğan and Taştan [5] recover the same ordering on the Turkish PISA 2018 subset, where a boosted regression tree edges out ridge, LASSO, bagging and random forests. For the post-pandemic window most relevant to us, Delprato [3] models low-performing students across ten Latin American systems in PISA 2022 and uses SHAP attributions to rank their determinants, foregrounding interpretation over any single accuracy number. Throughout, XGBoost, LightGBM and CatBoost [1, 10, 17] are the recurring strong baselines on this tabular regime, and SHAP values [11] the usual explanation layer.

Two regularities in this work frame our own contribution. First, reported performance sits in a narrow band, roughly AUC 0.78 to 0.84, or accuracy in the high seventies, largely irrespective of country, cycle, or the exact ensemble chosen. Second, the strong learners and the linear baselines are usually separated by a fraction of an AUC point, yet that gap is presented as a ranking rather than tested for significance, and even studies that expose it directly (linear models trailing on R^2 while nearly tying on AUC [4]) stop short of asking whether the difference is real or what would move it. The literature reports where the accuracy lands but rarely diagnoses why it lands there.

Accuracy ceilings, and what they are made of. That recurring ceiling near AUC 0.8 is usually treated as a fixed backdrop, a number to beat with a better learner or more tuning. We instead make it the object of study, and this is the paper's central methodological contribution. A plateau in predictive accuracy can have two very different causes. It may be a *feature ceiling*, in which a model has exhausted the signal in the feature families it was given but a richer family (here, the socioeconomic composition of a student's school) would move it; or it may be a *data ceiling*, in which the measurement instrument itself, the background questionnaire, cannot resolve risk any further and no feature engineering or model class will help. The two are indistinguishable from a single learning curve, and the reflex of tuning harder or adding capacity cannot separate them, because it holds the feature family fixed. We argue the distinction is nonetheless diagnosable, but only with *convergent evidence*: a feature ceiling is demonstrated when a new feature family lifts every model together, and a data ceiling is demonstrated when independent routes (a hand-crafted composition model, a design-appropriate multilevel model, and a direct school-questionnaire linkage) all plateau at the same value. We are not aware of a prior PISA study that draws this line explicitly, which is why the field's shared ~ 0.8 has gone largely uninterrogated.

Dataset shift. When the input distribution changes between two populations, a classifier trained to tell them apart quantifies the change, and a high domain-classifier AUC is direct evidence of covariate shift [18]. We use this as the central diagnostic for the 2018-to-2022 reversal rather than relying only on marginal score distributions, which cannot separate a merely lower-scoring cohort from one whose risk structure has moved.

Calibration, decision curves, and fairness. A screening probability must be calibrated to be actionable. Isotonic regression is a standard non-parametric post-hoc calibrator [27], and decision-curve analysis evaluates a screener by net benefit across operating thresholds rather than by accuracy at a single cut [22], which matters when prevalence is high. Group-fairness diagnostics such as demographic parity and equalized odds expose disparate error rates across protected groups [8]. We report all three, survey-weighted, a combination seldom applied together in the PISA setting.

Nested design and honest comparison. PISA students are nested in schools, so a random-intercept (multilevel) logistic model is the design-appropriate specification, and its intraclass correlation quantifies the between-school variance [6]. Finally, because repeated cross-validation folds share training data, naive fold-score intervals are anti-conservative; we use the Nadeau–Bengio corrected-resampled variance [13] and compare models with the corrected resampled t-test (in-sample) and the DeLong test on shared test predictions [2]. This is the machinery that turns the field's informal ranking of near-tied models into a tested claim about which gaps are real.

3. Data

3.1 Datasets

We use the student questionnaire files for all five PISA cycles from 2009 to 2022. The 2015/2018/2022 cycles ship as SPSS `.sav`; 2009 and 2012 are fixed-width ASCII with SPSS control files, which we parse with a purpose-built reader. After harmonising feature names across cycles, the Albanian longitudinal file holds **27,042** students. A comparison set of nine countries (Albania, North Macedonia, Montenegro, Serbia and Bulgaria as Balkan peers; Estonia and Finland as top performers; and Mexico and Colombia as GDP-matched economies) holds **323,121** students. Raw files are multi-gigabyte and not redistributed; processed slices are regenerated by the pipeline.

3.2 Target variable

A student is *at risk* if their mathematics score falls below the exact OECD Level 2 boundary of 420.07 points. Because each score arrives as ten plausible values $\mathrm{PV}_1, \ldots, \mathrm{PV}_{10}$, the model-comparison target is the point (PV-mean) indicator,

$$\texttt{AT\_RISK} = \begin{cases} 1 & \text{if } \frac{1}{10}\sum_{j=1}^{10} \mathrm{PV}_j < 420.07 \quad \text{(below Level 2)} \\ 0 & \text{otherwise (proficient)}, \end{cases}$$

while the final estimates use per-plausible-value targets combined by Rubin's rules. In the 2022 crisis cohort the class balance is 75.3% at risk versus 24.7% proficient.

3.3 Features

The core set is thirteen background variables with high cross-cycle availability: ESCS (the OECD socioeconomic index), home possessions, parental education (HISCED) and occupation (HISEI), gender, grade repetition, immigration status, school belonging, teacher support in mathematics, math anxiety, grade relative to modal, and home/school ICT resources. Composite and interaction features (a standardised SES composite, a material-deficit rank, digital-readiness indices, and centred interaction terms) are derived *inside* the cross-validation pipeline on the training fold only, so their scaling and rank statistics never leak from the test fold.

3.4 Data dictionary and provenance

Table 1 names each analytic variable, its plain-English meaning, the direction of its scale, and the cycles in which Albania actually carries it. The OECD background indices (ESCS, HOMEPOS, BELONG, ...) are standardised to an OECD mean 0 / SD 1, so a value of −0.75 reads directly as "0.75 SD below the OECD student average" before any modelling.

3.5 Survey-weighted descriptive summary

Table 2 is the population-level "know your data" view of the 2022 crisis cohort: weighted mean and SD, the raw range, the weighted median, and how much is missing. Two facts stand out. First, Albania sits far below the OECD SES origin (ESCS −0.75, HOMEPOS −0.87). Second, the attitudinal indices carry heavy missingness (34–40% for teacher support and math anxiety), which caps how much any model leaning on them can contribute, a caveat that recurs downstream.

3.6 Which features separate at-risk students?

Ranking predictors by the *strength* of their weighted association with low proficiency, not just by significance (since at $n \approx 6{,}000$ almost everything is "significant"), gives Table 3. Numeric features use weighted Cohen's d (at-risk vs. proficient; sign shows direction), categoricals use Cramér's V. Home possessions, ESCS and *math anxiety* are the strongest separators ($|d| \approx 0.5$, medium); the demographic categoricals are trivially small ($V < 0.1$), and immigrant status is not even significant ($p \approx 0.16$). Risk here is an SES-and-affect story far more than a demographic one, and it sets the bar a later feature-importance analysis (§5) must beat.

Table 1: Analytic data dictionary. OECD indices are standardised (mean 0, SD 1 across OECD). "Cycles" lists the Albanian cycles with non-missing values.

Variable	Meaning	Direction	Cycles
MATH_PV_MEAN	Math score, mean of 10 plausible values	higher = better	all 5
AT_RISK	Low proficiency: math < 420 (below Level 2)	1 = at risk	all 5
ESCS	Economic, social & cultural status index	higher = advantaged	'09,'18,'22
HOMEPOS	Home possessions (material + educational)	higher = more	all but '15
HISCED	Highest parental education (ISCED level)	higher = more	'09,'18,'22
HISEI	Highest parental occupation status (ISEI)	higher = higher	'09,'18,'22
BELONG	Sense of belonging at school	higher = more	all but '15
TEACHSUP	Perceived teacher support in maths	higher = more	'12,'18,'22
ANXMAT	Mathematics anxiety	higher = more anxious	'12,'22
REPEAT	Has repeated a grade	1 = repeated	all but '15
IMMIG	Immigrant background	1 = 1st/2nd gen	all but '15
GRADE	Grade relative to modal grade	0 = modal	all 5
GENDER	Student gender	coded 0/1	all 5

Table 2: Survey-weighted descriptive summary, Albania 2022. Mean/SD/median use final student weights; min/max are raw sample extremes.

Variable	n	%miss	w-mean	w-SD	median	range
MATH_PV_MEAN	6129	0.0	368.2	79.0	357.5	142–693
ESCS	5575	9.0	−0.75	1.09	−0.77	−5.2–2.8
HOMEPOS	5699	7.0	−0.87	1.02	−0.89	−6.3–5.4
HISCED	4603	24.9	5.60	2.51	5.0	1–10
HISEI	4773	22.1	45.2	23.5	31.7	11–89
BELONG	4743	22.6	0.25	1.05	0.14	−3.3–2.8
TEACHSUP	4002	34.7	0.41	1.25	0.82	−2.9–1.6
ANXMAT	3674	40.1	0.01	1.17	0.05	−2.4–2.6
GRADE	4734	22.8	−0.04	0.35	0.0	−3–2

3.7 Cross-country comparability

Before ranking countries we ask whether the 2022 samples are comparable (Table 4). They are broadly matched in size and ESCS coverage, so the ranking that follows is not an artefact of one country being thinly or selectively measured. The table also states the paper's puzzle numerically: Colombia and Mexico sit at a *lower* mean ESCS than Albania yet post *lower* low-proficiency rates. SES level alone therefore does not explain Albania's position; the gradient's slope, examined in §5, does the work.

3.8 The socioeconomic gradient as a probability curve

Figure 1 shows the continuous gradient, $P(\text{low proficiency} \mid \text{ESCS})$, as a descriptive weighted-logistic curve with the weighted at-risk rate per SES decile overlaid, for 2018 versus 2022. The 2022 curve sits far above 2018 at *every* SES level: the crisis raised risk across the board, not only for the disadvantaged. The gradient (slope) persists, so SES still matters, but a pure

Table 3: Weighted effect sizes vs. the at-risk target, Albania 2022, ranked by magnitude. Numeric: Cohen's d; categorical: Cramér's V.

Feature	Measure	Value	p
HOMEPOS	Cohen's d	−0.504	n/a
ESCS	Cohen's d	−0.496	n/a
ANXMAT	Cohen's d	+0.486	n/a
HISEI	Cohen's d	−0.455	n/a
BELONG	Cohen's d	−0.231	n/a
TEACHSUP	Cohen's d	−0.141	n/a
REPEAT	Cramér's V	0.104	< 0.001
GENDER	Cramér's V	0.074	< 0.001
GRADE	Cohen's d	−0.044	n/a
HISCED	Cohen's d	−0.040	n/a
IMMIG	Cramér's V	0.027	0.161

Table 4: Sample composition and comparability, 2022 (survey-weighted), sorted by at-risk rate. Colombia and Mexico are poorer yet less at-risk than Albania.

Country	N	mean score	mean ESCS	at-risk %	ESCS %miss
Albania	6129	368.2	−0.75	75.3	9.0
Colombia	7804	382.7	−1.07	72.6	4.4
Mexico	6288	395.0	−0.95	67.2	0.3
North Macedonia	6610	388.6	−0.28	67.1	3.0
Montenegro	5793	405.6	−0.21	60.2	1.5
Bulgaria	6107	417.3	−0.27	54.0	4.0
Serbia	6413	439.9	−0.20	43.2	1.0
Finland	10239	484.1	+0.26	24.2	3.0
Estonia	6392	509.9	+0.15	13.8	1.6

"poverty shock" story would predict a steepening the data only partly show, consistent with the feature-drift result in §5. This is a single-predictor descriptive gradient, distinct from the multivariable models below.

3.9 Design-based estimates

The 2015/2018/2022 cycles carry 80 Fay balanced-repeated-replication weights; we combine the BRR sampling variance with the between-plausible-value variance to report the standard error the OECD manual prescribes. The 2009/2012 fixed-width cycles do not carry replicate weights and fall back to an imputation-only standard error, which we flag. Table 5 gives Albania's weighted at-risk trajectory with these design-based standard errors: a V-shape that falls from 68% (2009) to 42% (2018) and then snaps back to 74% in 2022 (Figure 2).

3.10 A verified data gap

Albania genuinely lacks the ESCS index in 2012 and 2015 (an OECD coverage gap confirmed against the source files, not a processing bug), whereas all peer countries have full 2015 background data. This limits Albania's *own* cross-cycle socioeconomic comparison, but it leaves the 2018-versus-2022 shift analysis, in which both cycles are fully covered, unaffected.

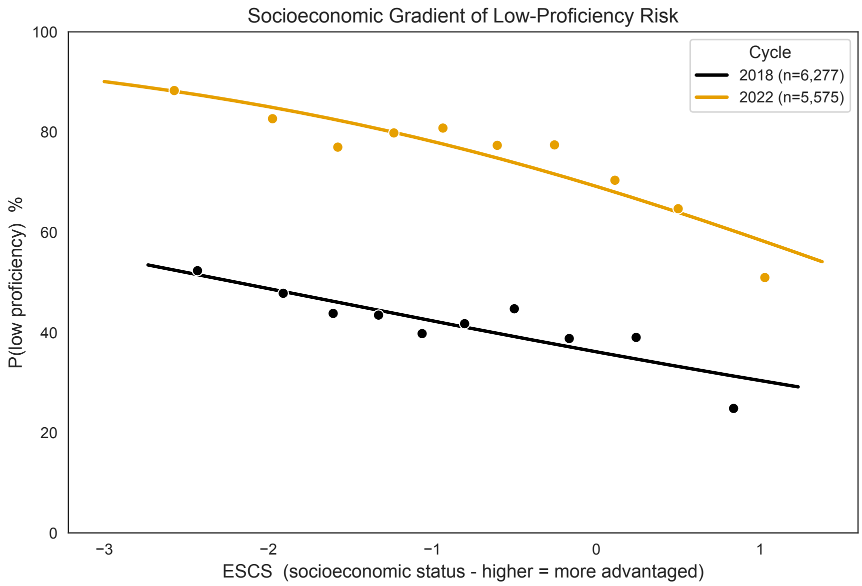

Figure 1: Descriptive socioeconomic gradient of low-proficiency risk, 2018 vs. 2022. Markers are weighted at-risk rates per ESCS decile; lines are weighted logistic fits. The 2022 curve is shifted up at every SES level.

Table 5: Albania weighted at-risk mathematics rate (below Level 2) with design-based standard errors. BRR available only for the SPSS cycles.

Cycle	At-risk	SE	95% CI	BRR
2009	0.677	0.0076	0.663–0.692	n/a
2012	0.607	0.0051	0.597–0.617	n/a
2015	0.533	0.0108	0.512–0.554	✓
2018	0.424	0.0088	0.407–0.441	✓
2022	**0.739**	**0.0041**	**0.731–0.747**	✓

4. Methods

4.1 Survey-correct estimation

Every descriptive statistic and every model metric is weighted by the final student weight W_{FSTUWT}. Where a model is fit, weights are routed to the estimator's own `sample_weight` (correcting a subtle bug in which scaler-wrapped models, logistic regression, SVM, KNN, MLP, silently fell back to an unweighted fit because the weight was passed to the wrong pipeline step). Class-conditional statistics rescale the population-scale weights to the effective sample size before, e.g., a χ^2 test, so significance is not inflated by the weight magnitudes.

4.2 Leakage-safe pipeline

Each fold's pipeline is: (1) an engineered-feature builder that learns component means, standard deviations, and the home-possessions rank curve on the training fold and replays them on the test fold; (2) median imputation fit on the training fold; (3) the estimator. Interaction terms are built from mean-centred components so a product is not confounded with its main effects. School-composition features are survey-weighted, *leave-one-out* school means (a student's own value is excluded from their school mean), and a fold-safe re-computation that recomputes the

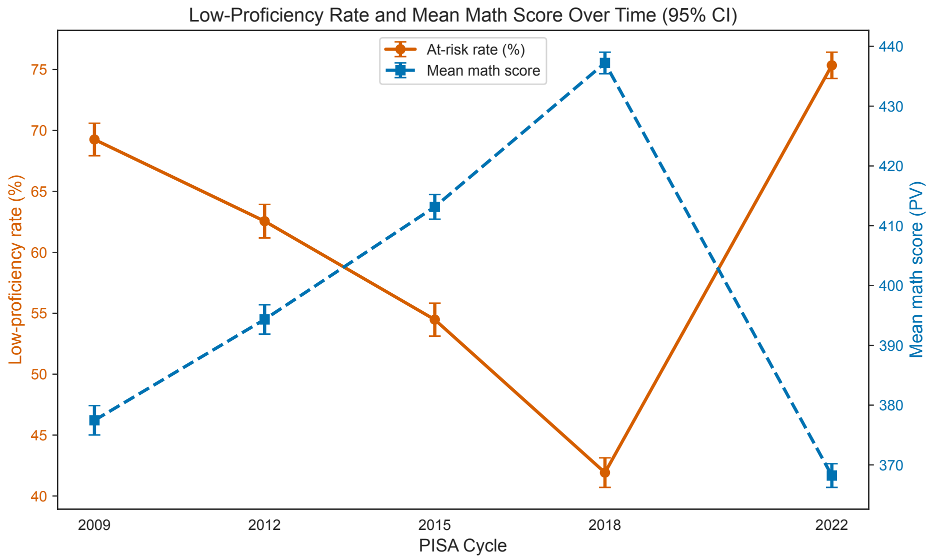

Figure 2: Albania's weighted below-Level-2 mathematics rate, 2009–2022. The 2018 low (42%) and the 2022 spike (74%) bracket the structural break this paper explains.

means from the training fold only reproduces an identical effect, so the lift is compositional, not leakage.

4.3 Evaluation and comparison

Models are scored with repeated stratified 5×4 cross-validation, weighted ROC and PR AUC, F1-macro, and MCC. Fold-score confidence intervals use the Nadeau–Bengio corrected-resampled variance, which replaces the naive $1/k$ term with $1/k + n_{\text{test}}/n_{\text{train}}$ to account for the correlation between folds that share training data. Model-vs-model differences are tested directly: the corrected resampled t-test on paired per-fold AUCs (cross-validation) and the DeLong test on shared test-set predictions (out-of-sample). This turns a ranking table into a claim about which gaps are real.

4.4 Covariate-shift diagnostic

To quantify the 2018→2022 change we train a domain classifier to distinguish the two cohorts from background features alone, with weighted subsampling, pooled-median imputation, and missingness indicators; a high AUC certifies that the input distribution, not just the score level, moved. We validate the diagnostic on a null (same cohort split at random → AUC ≈ 0.5).

4.5 Explanation, calibration, fairness, forecast

We compute TreeSHAP global and per-student attributions on the school-context model, and partial-dependence / centred-ICE curves for the top drivers. For deployment we add out-of-fold isotonic calibration and tune the decision threshold on calibrated out-of-fold predictions to maximise weighted MCC. The fairness audit reports weighted demographic parity, equal-opportunity (TPR) and false-positive-rate gaps across gender, SES quintile, and immigration status, plus a threshold sweep. A random-intercept logistic model provides the design-appropriate specification and the between-school intraclass correlation. Finally, a scenario Monte-Carlo forecast for 2026 propagates each cycle's design-based standard error through a weighted trend fit; with only five cycles and a structural break we report scenarios and their intervals, not a point prediction.

4.6 Robustness and extension analyses

Five further analyses probe the main claims. (i) *Cross-country ceiling replication*: the student-only versus school-context comparison is re-run identically in all nine systems, and each system's between-school ICC is read from its random-intercept model, to test whether the feature ceiling and its size generalise. (ii) *Modality ablation*: PISA's computer-based cognitive-process indicators are added to the school-context model and split into cognitive-process and questionnaire-process blocks, so an endogenous within-test signal can be separated from data available at screening time. (iii) *Fairness mitigation*: group-specific decision thresholds are fit on calibrated out-of-fold scores to equalise the per-quintile false-positive rate (equalized-odds post-processing), tracing the accuracy–equity frontier. (iv) *Partial identification*: given the 79% coverage index, worst-case and monotone Manski bounds on the population at-risk rate are computed, the monotone bound assuming the out-of-frame group is no better off than the assessed. (v) *Difference-in-differences*: a design-based DiD contrasts the earthquake-affected central region with the rest of the country across 2018 and 2022, with a 2015→2018 placebo to test parallel trends; weights, plausible values, and replicate-weight standard errors are carried throughout.

4.7 Reproducibility and engineering

Logic lives in unit-tested modules (177 tests on synthetic fixtures, covering the weighted statistics, Rubin's rules, the leakage-safe transformer, the BRR+PV variance, the Manski bounds and group-threshold mitigation, and the Nadeau–Bengio / DeLong maths); notebooks are narrative layers over them. On macOS the boosters (Homebrew `libomp`) and scikit-learn (bundled OpenMP) can abort a process if their runtimes collide; we isolate each model fit in a fresh subprocess that imports the boosters before scikit-learn, which resolved an import-order segfault and returned LightGBM to the comparison.

5. Results

5.1 The 2022 collapse is a structural shift

A domain classifier separates 2018 from 2022 Albanian students with weighted AUC **0.98** on background features alone. The 2022 cohort is not simply a lower-scoring version of 2018 along a stable structure; the input distribution itself moved. The most-shifted variables (Figure 3) are teacher support in mathematics (standardised mean difference -0.31) and home possessions ($+0.27$), with school belonging, grade repetition, and ESCS following. That teacher support fell while reported home possessions rose is consistent with a disruption to the *schooling process* rather than a household-wealth shock, and it foreshadows the compositional result below. Out-of-sample the shift is detectable but not catastrophic: models trained on 2009–2018 still transfer to 2022 at AUC ≈ 0.67 (covariate shift without total collapse).

5.2 An accuracy ceiling, and its cause

On the thirteen student features, twelve estimators cluster tightly (Figure 4). CatBoost leads at weighted CV AUC 0.731, but a paired Nadeau–Bengio test finds the top three statistically *tied*: CatBoost $\approx$ logistic regression ($p = 0.50$) $\approx$ gradient boosting ($p = 0.37$): a transparent linear model matches the best boosters. Nested-CV Optuna tuning of the top three yields no significant gain (CatBoost $+0.002$, $p = 0.81$; GBM $+0.004$, $p = 0.45$; LR -0.001, $p = 0.43$). This looks like a data ceiling.

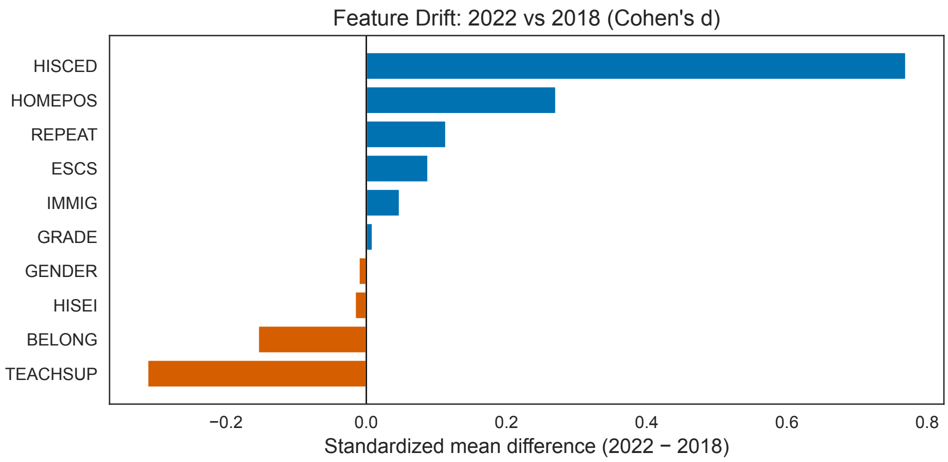

Figure 3: Standardised 2018→2022 drift of Albanian background features. Teacher support and home possessions move most; the domain classifier that separates the cohorts reaches AUC 0.98.

Table 6: Albania 2022, weighted 5×4 CV, *with* school context. ΔAUC is the lift over the student-only baseline.

Model	AUC	ΔAUC	PR-AUC	F1	MCC
CatBoost	**0.783**	+0.052	0.907	0.693	0.393
Gradient Boosting	0.777	+0.050	0.905	0.665	0.365
LightGBM	0.775	+0.060	0.902	0.688	0.381
Random Forest	0.769	+0.060	0.895	0.644	0.345
Extra Trees	0.758	+0.051	0.884	0.641	0.332
XGBoost	0.756	+0.058	0.892	0.657	0.333
Logistic Regression	0.754	+0.027	0.897	0.649	0.332
SVM (RBF)	0.749	+0.037	0.894	0.609	0.282
MLP (128-64)	0.743	+0.034	0.884	0.620	0.295
Naive Bayes	0.701	+0.019	0.856	0.610	0.263
KNN	0.700	+0.036	0.861	0.613	0.257
Decision Tree	0.609	+0.041	0.797	0.608	0.217

It is not; it is a *feature* ceiling. Adding survey-weighted, leave-one-out **school-mean** ESCS, home possessions, anxiety, and teacher support, plus cohort size, lifts every model by ∼0.05 AUC (Table 6): CatBoost 0.731 → **0.783**, and the effect is identical under fold-safe re-computation. Crucially, adding school context *breaks the tie*: the boosters now beat logistic regression significantly (GBM vs LR $p = 0.001$), because they exploit a student × school-composition interaction a linear model cannot. SVM and MLP, added for completeness, land mid-table: the podium is a booster story, not an artefact of omitting kernel or neural learners.

5.3 ∼0.78 is a genuine data ceiling

Three independent routes converge on ∼0.78. (i) The composition booster above plateaus there. (ii) A random-intercept logistic model, the design-appropriate specification, *matches* the hand-crafted school-mean booster on identical folds (CV AUC ≈ 0.78) and reports a school intraclass correlation ICC ≈ 0.31: about a third of risk variance is *between* schools and barely touched by student features. (iii) Linking the PISA school questionnaire (principal-reported resources, staff, class size, leadership) directly onto students adds no significant signal beyond composition (best +0.004 AUC, all $p > 0.24$), because school-mean SES already proxies the school inputs.

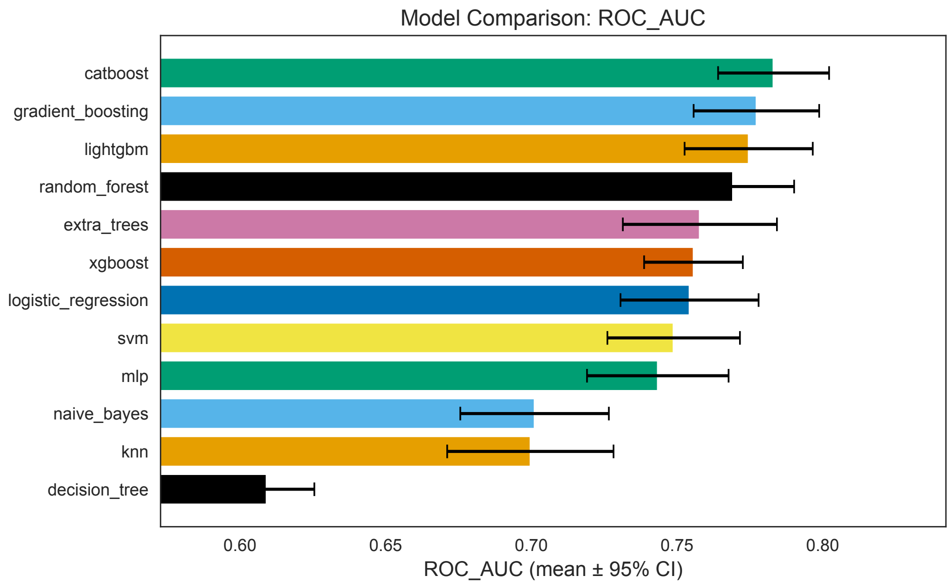

Figure 4: Weighted CV AUC with Nadeau–Bengio intervals. Boosters form a top band once school context is included; the interpretable logistic-regression baseline gains least because it cannot use the student $\times$ school interaction.

Table 7: Student-only vs. school-context AUC across the nine 2022 systems. The composition lift is significant everywhere but Estonia and scales with the between-school ICC ($r \approx 0.80$ between ICC and school-context AUC).

Country	Group	Base	School	Lift	ICC
Albania (ALB)	Balkan	0.714	0.773	+0.059	0.31
North Macedonia (MKD)	Balkan	0.769	0.835	+0.066	0.40
Montenegro (MNE)	Balkan	0.734	0.817	+0.083	0.34
Serbia (SRB)	Balkan	0.745	0.805	+0.060	0.36
Bulgaria (BGR)	Balkan	0.791	0.857	+0.066	0.56
Colombia (COL)	GDP-matched	0.834	0.863	+0.028	0.45
Mexico (MEX)	GDP-matched	0.749	0.794	+0.045	0.31
Estonia (EST)	Top performer	0.729	0.752	+0.022	0.21
Finland (FIN)	Top performer	0.781	0.790	+0.008	0.10

The ceiling is a property of what background questionnaires can measure, not of tuning or model capacity. An ensemble stack of the four boosters confirms this: it reaches AUC 0.786 vs single CatBoost 0.783 (+0.003, $p = 0.03$): statistically significant, practically nil, and it *regresses* MCC and F1 at $\sim$19$\times$ the compute. Single CatBoost stays the deployable headline.

The ceiling is not an Albanian idiosyncrasy. Replaying the identical student-only versus school-context comparison across all nine systems (Table 7) reproduces the pattern almost everywhere: the composition lift is significant in eight of nine countries (the exception is low-inequality Estonia, $p = 0.09$), and its *size* tracks how much risk variance sits between schools. The correlation between a system's school intraclass correlation and its school-context AUC is $r \approx 0.80$; where schools are socioeconomically segregated (Bulgaria ICC 0.56, North Macedonia 0.40) composition buys the most, and where they are equitable (Finland ICC 0.10) it buys almost nothing (+0.008 AUC). The feature-versus-data ceiling is thus a general property of background-questionnaire prediction, and the amount a system can move it is set by its own between-school structure rather than by the learner.

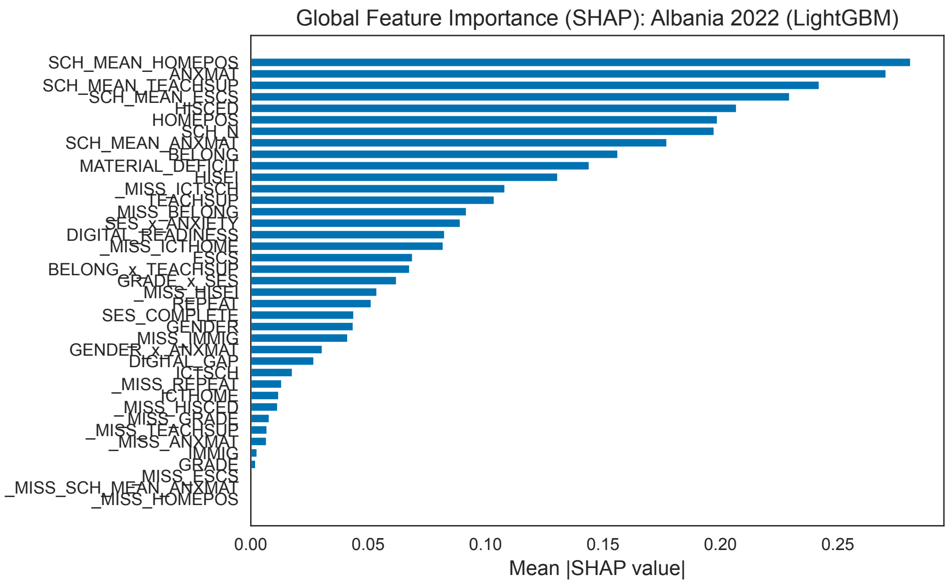

Figure 5: Global SHAP importance (school-context model). Four of the top seven drivers are school-level; math anxiety is the strongest individual factor.

5.4 The ceiling is a measurement limit, not a modelling one

If the ∼0.78 plateau is a property of what the *background* questionnaire can see, then a genuinely richer modality should move it, whereas more of the same questionnaire should not. Both hold. Adding PISA's computer-based cognitive-process data (the 15 timing-, action-, and effort-derived indicators from the assessment delivery itself) lifts the school-context model from AUC 0.773 to **0.858** (+0.085, $p < 0.001$). But almost all of that gain is the cognitive-process block: cognitive features alone reproduce it (+0.078), while the non-cognitive *questionnaire*-process features add only +0.011 ($p = 0.02$). The cognitive-process signal is endogenous to the outcome, it is measured *during* the same test whose result we predict, so it inflates apparent accuracy without being available at screening time. The deployable ceiling, over data a school could hold *before* a student sits the assessment, therefore stays at ∼0.78. Richer data can move the plateau, but only data that is not a downstream trace of the score itself, which confirms the ∼0.78 figure as a measurement limit rather than a modelling one.

5.5 Drivers: anxiety and school composition, shared with peers

On the school-context model, four of the top seven SHAP drivers are school-level (Figure 5): school-mean home possessions leads, followed by individual math anxiety, school-mean teacher support, and school-mean ESCS. The model reads a student's risk more from the socioeconomic *composition* of their school than from their own resources; this is the compositional effect made explicit. Immigration status is negligible. A feature × country SHAP rank matrix (Figure 6) shows these drivers are *shared*: math anxiety is a top-3 driver almost everywhere (including Estonia), and school composition dominates most systems. What is distinctive about Albania is the *level and breadth* of low proficiency, not the mechanism.

5.6 A broad-based, flat-gradient crisis

Per-country school-context models (Figure 7) place Albania at the highest at-risk rate (75%) but one of the *lowest* model AUCs (0.77): with three in four students at risk there is little

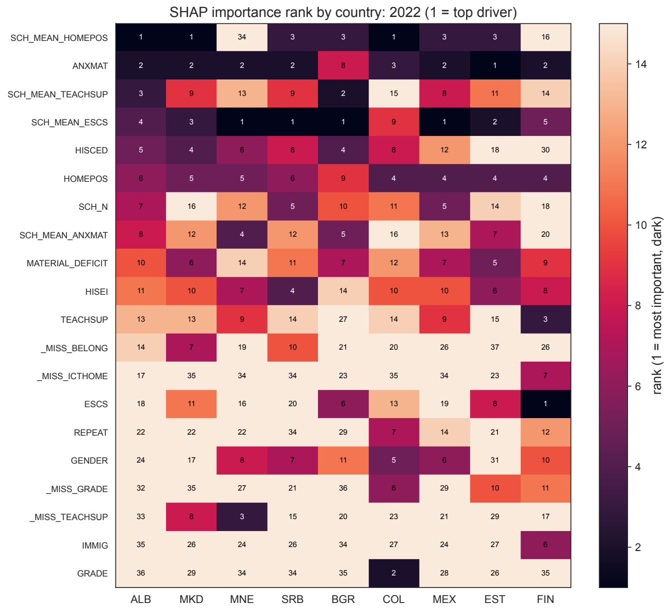

Figure 6: SHAP driver ranks by country. Math anxiety and school composition recur across systems; Albania mirrors its peers in mechanism.

contrast to separate. Albania also has the **flattest** SES gradient of the nine countries (∼17 score points per SD), a floor effect: even advantaged Albanian students score near 369, close to the line. Estonia and Finland have *steeper* gradients (∼38–40) yet far higher levels, because there is more score range for SES to predict. The policy reading is direct: a flat gradient plus school-composition dominance means SES-targeted transfers alone miss most at-risk Albanian students; system-wide levers (teacher support, math-anxiety reduction, raising the floor across all schools) are required.

5.7 A calibrated, fair-audited screener

Raw CatBoost probabilities are sharp but mis-calibrated (weighted ECE ≈ 0.12); out-of-fold isotonic calibration cuts this to ≈ 0.01 (Figure 8), and decision-curve analysis shows the model beats screen-everyone / screen-no-one across the selective threshold band, the relevant test at 75% prevalence. The fairness audit is less comfortable (Figure 9): at a fixed threshold the screener over-flags the poorest SES quintile (false-positive rate ≈ 0.86 vs ≈ 0.24 for the richest; parity gap 0.48, FPR gap 0.63). Because school-mean SES *is* SES, making it a top feature doubly encodes the gradient; adding school context shrinks the gender gap but leaves the SES gap intact. This is an equalized-odds concern.

The gap is, however, fixable at the operating point. Post-processing the calibrated scores with *group-specific* thresholds (a per-quintile cut chosen to equalise the false-positive rate, the equalized-odds remedy) collapses the SES FPR gap from 0.63 to **0.003** while holding the overall false-positive rate fixed. It is not free: matching the poorest quintile's threshold to the rest trades recall (overall $0.81 \to 0.75$) and accuracy ($0.75 \to 0.70$) for equity, the familiar fairness–accuracy tension made explicit rather than hidden. We therefore ship the single-threshold screener as the default but expose the group-threshold frontier, and frame either as a tool to *widen* support, never to ration it.

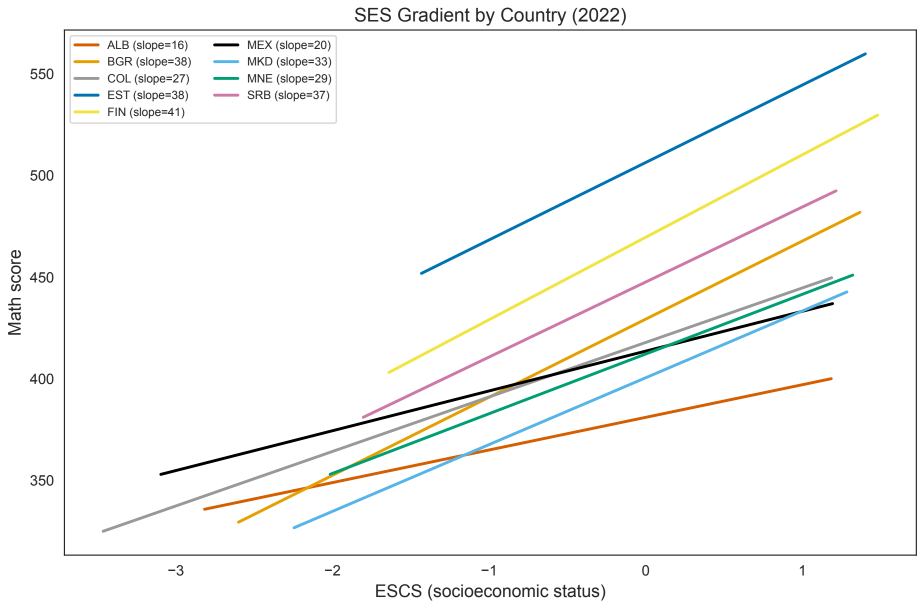

Figure 7: SES gradient of mathematics score, 2022. Albania's slope is the flattest of the nine countries, a floor effect, not a steep advantage gap.

5.8 Is the 75% an artefact of coverage? Partial-identification bounds

Albania met only part of PISA's sampling standards in 2022, and its Coverage Index 3 is 0.79: roughly a fifth of the 15-year-old population is out of frame, mostly adolescents who have left or never entered school. Because that missing fifth is plausibly *worse* off, the headline rate could in principle be an underestimate, so we ask what the data alone can guarantee without assuming anything about the uncovered. The assessed sample's weighted at-risk rate is 0.739 (design-based 95% CI 0.731–0.747). Worst-case Manski bounds, letting every uncovered student take any value, give [0.584, 0.794]; the more defensible monotone bound, assuming the out-of-frame group is at least as at-risk as those assessed, gives [**0.739**, **0.794**]. Under either identifying assumption the 2026-relevant crisis level stays far above Albania's 2018 rate of 0.42: coverage counsels caution on the exact percentage but cannot explain the collapse away, and if anything the true breadth is wider than the measured ∼0.75.

5.9 Is the collapse local to the earthquake? A difference-in-differences

The 2019 Durrës–Tirana earthquake is the one shock Albania did not share with its peers, so it is tempting to credit the collapse to it. A design-based difference-in-differences tests this directly, comparing the earthquake-affected central region against the rest of the country across the 2018 and 2022 cycles (final weights, plausible values by Rubin's rules, 80 replicate weights). The affected region's at-risk rate did rise 4.7 points more than the control's (DiD = +0.047, $p < 0.001$), which in isolation looks like a local earthquake effect. It is not identified: a *placebo* DiD on the pre-earthquake 2015→2018 window is itself large and significant with the opposite sign (−0.070, $p < 0.001$), so the two regions were already on diverging trajectories and the parallel-trends assumption fails. The credible reading is the one the magnitudes give directly, that both regions collapsed by roughly the same enormous amount (treated +0.34, control +0.30): the crisis is *national*, not concentrated at the epicentre. This is a well-identified *null* for a local causal story and independent corroboration of the broad-based, system-wide account, the pandemic-era disruption to schooling reached every region, and no single-region natural experiment can carry the explanation.

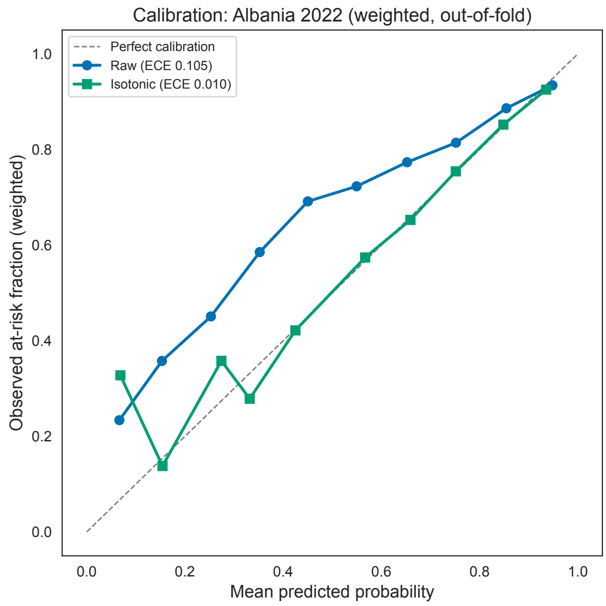

Figure 8: Reliability curve before/after out-of-fold isotonic calibration; weighted ECE falls from ≈ 0.12 to ≈ 0.01.

5.10 Forecast for 2026

With only five cycles and a 2022 structural break, a point forecast is indefensible, so we report a Monte-Carlo scenario band with design-based standard errors propagated. The honest message is its width: the defensible zone for the 2026 below-Level-2 rate runs from persistence of the crisis (median 0.74, 90% PI 0.66–0.82) through partial reversion (0.47, 0.43–0.52) to a full pre-COVID recovery (0.21, 0.18–0.24), and even the optimistic tail sits above Albania's 2018 low of 42%. We treat the forecast as context for the structural findings above, not as a headline result.

6. The Economic and Structural Context of the Collapse

The statistical results locate the collapse in time and shape but not in cause; the economic and structural history of Albania between the 2018 and 2022 assessments supplies the missing narrative. PISA 2018 was fielded in early 2018 and PISA 2022 in 2022, so any explanation must point to forces that acted in that four-year window. Two shocks did, and only one of them was shared with Albania's peers.

Two compounding shocks hit the schooling window. Albania entered the pandemic already weakened by a natural disaster that its comparison countries did not face. On 26 November 2019 a magnitude 6.4 earthquake struck the Durrës and Tirana region, the strongest to hit the country in more than four decades; it killed 51 people, injured about 3,000, and damaged or destroyed housing and schools across roughly a dozen cities in the most populous part of Albania [7]. Barely three months later, COVID-19 closed schools nationwide from March 2020, and Albania recorded one of the longer cumulative closure durations in the region before instruction fully resumed [20, 25]. The comparison economies in this study lived through the pandemic, but none of them absorbed a major earthquake in the same window. Notably, the OECD's own country note attributes Albania's 2022 decline to exactly this pairing, an end-of-2019 earthquake that damaged schools and houses across twelve cities, compounded by the pandemic [16]. Albania therefore experienced a *double* disruption to the continuity of

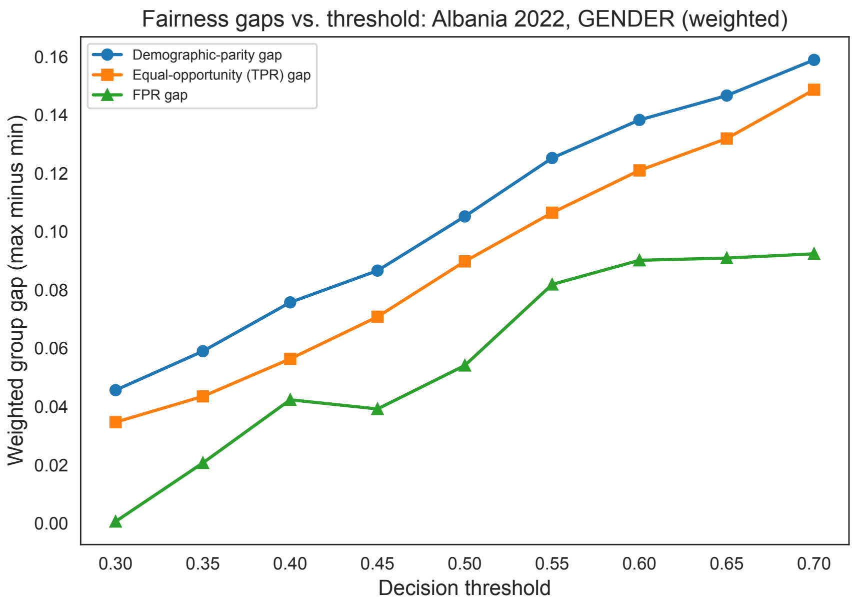

Figure 9: Weighted fairness gaps by decision threshold (gender shown). Gaps shift with the operating point rather than vanishing, mitigation needs constraints, not a knob.

schooling precisely between the two assessments, which is where a structural break would have to originate.

The shock was to schooling, not to household wealth, and the data show it. If the 2022 collapse were a household-income shock, the socioeconomic markers in the questionnaire would have fallen; instead, the school-side marker fell while the household marker held. The covariate-shift analysis (Section 5) found that the variable which moved most from 2018 to 2022 was teacher support in mathematics, down by roughly a third of a standard deviation, while reported home possessions actually rose. A demand-side poverty shock would depress home possessions; the observed pattern instead points to a supply-side degradation of instruction, the signature of closed and damaged schools and of teaching capacity stretched thin. This interpretation is consistent with the macroeconomic record: Albania's economy contracted in 2020 and rebounded strongly the next year, with World Bank figures placing the contraction at roughly three to four percent and 2021 growth near nine percent as tourism recovered [23], while household consumption was cushioned throughout by diaspora remittances worth about nine percent of GDP [24]. Family assets did not crater, which is why home possessions did not; the damage concentrated in the public delivery of schooling.

Structural weaknesses turned a temporary shock into a lasting shift. A closure need not become a permanent loss, but three structural features of Albania's system made it likely to. First, remote learning depended on infrastructure that many households lacked: home internet and device access, especially in rural areas, left a substantial share of students unable to follow online instruction, so televised and online lessons reached pupils unevenly [21, 25]. Second, Albania spends a comparatively low share of GDP on education, on the order of three to four percent, below the world average of about 4.6 percent and below European Union norms [26], leaving little slack to mount an intensive catch-up response. Third, Albania has for years experienced heavy emigration of working-age and skilled adults, including teachers; the 2023 census confirmed that the resident population had fallen by roughly 14 percent, about 420,000 people, since 2011, to some 2.4 million [9]. This brain drain thins instructional capacity at precisely the moment recovery demands more of it. Together these conditions explain why the disruption still registered, undiminished, in the 2022 scores rather than bouncing back.

The economic reading matches the statistical shape of the crisis. A system-wide capacity shock predicts a broad-based decline, and that is what the gradient analysis found. Because closures, damaged buildings, and thinned teaching capacity act on all students in a school rather than only its poorest, such a shock should depress even advantaged students and flatten the socioeconomic gradient. Albania in 2022 has the flattest SES gradient of the nine countries and a floor so low that even its advantaged students score near the proficiency line (Section 5); its mean mathematics score fell from 437 in 2018 to 368 in 2022, a drop of about 69 points [16]. A shock concentrated in household poverty would have steepened the gradient, not flattened it. A system-wide shock also predicts a *geographically* broad decline, and a difference-in-differences confirms it: the mathematics collapse is no larger in the earthquake-affected central region than elsewhere once a placebo exposes diverging pre-trends (Section 5), so the 2019 earthquake compounded but did not localise the loss, the decisive disruption was the national, pandemic-era interruption of schooling. The convergence of the economic history and the statistical shape is the central insight of this section: the collapse behaves like a supply-side disruption to schooling, not a demand-side collapse in family means.

A caveat on measured effort. One qualification tempers the magnitude, though not the direction, of the drop. The OECD country note reports that Albanian students spent less effort on the 2022 assessment than in 2018 and shows signs of lower test engagement [16], so part of the measured score fall may reflect test-taking behaviour rather than learning alone. This does not undo the structural reading. Our central shift diagnostic (Section 5) is trained on background questionnaire responses, not on the scores themselves, and still separates the 2018 and 2022 cohorts at AUC 0.98; the change in *who* the students are is real independent of how hard they tried on test day.

A caveat on sample coverage. A second qualification concerns who the 2022 sample represents. The OECD flags that one or more of its sampling standards were not met for Albania in PISA 2022, and Albania's Coverage Index 3, the share of the national 15-year-old population that the assessed sample represents, is about 79%: the 6,129 students tested in 274 schools stand in for roughly 28,400 of an estimated 36,000 15-year-olds [15, 16]. The out-of-frame fifth is disproportionately made up of adolescents who have left school or were never enrolled, who as a group would be expected to perform below those still in the system. If anything, then, this exclusion biases the measured at-risk rate *downward*, so the true breadth of low proficiency is likely wider than the already severe 75% we report, and the collapse is not an artefact of sweeping in weaker students. We make this precise with partial-identification bounds (Section 5): even the worst-case Manski interval that makes no assumption about the uncovered fifth is [0.58, 0.79], and the monotone bound that treats them as no better off than the assessed is [0.74, 0.79], so under any defensible assumption the crisis rate stays far above Albania's 2018 level of 42%. Two design points guard the central comparison further. Coverage was of comparable order in 2018, so the 2018-to-2022 contrast is not driven by a coverage change; and our structural-shift diagnostic conditions on background questionnaire responses within the assessed population rather than on population totals, so it is insensitive to the level of coverage even if it shifted. Coverage does, however, counsel caution in reading the absolute rate as a precise population figure, which is one more reason we lean on the shift in risk structure rather than on the headline percentage alone.

So what this implies for policy. Reading the collapse as broad-based and supply-side changes the policy prescription. If the problem were concentrated household poverty, socioeconomically targeted transfers would reach most at-risk students; because it is a flat, system-wide degradation of schooling capacity, such transfers alone would miss them. The levers the evidence points to instead are those that rebuild the delivery of instruction: retaining and paying teachers to slow the brain drain, completing physical reconstruction of damaged schools, funding sustained catch-up teaching, and addressing the two individual factors the model ranks highest, math anxiety and the quality of teacher support. These are the same system-wide levers that the shared, broad-based driver structure in Section 5 already implied, now grounded in the country's economic history.

7. Discussion

What the shift means. The single most important result is that the 2022 collapse is structural: a classifier tells 2018 from 2022 Albanian students almost perfectly from background data, and the variables that moved most are process variables: teacher support fell while reported home possessions rose. This is not the signature of a household-wealth shock; it is consistent with a disruption to schooling itself (the pandemic and its long tail), and it explains why models trained on the pre-crisis trajectory transfer only moderately to 2022. A "weaker cohort" framing would misdirect policy toward uniform remediation; a structural-shift framing points at the school process.

Why the ceiling matters. The instinct on hitting a plateau is to tune harder or add capacity. Here that instinct is wrong twice over. Tuning did nothing; capacity (a neural net, a kernel machine, a four-model stack) did nothing. The missing ingredient was a *feature family*, school socioeconomic composition, and once added, the plateau moves and the model class that wins changes. But the new plateau at $\sim$0.78 is then real, corroborated by a multilevel ICC of $\approx$0.31 and by a direct school-questionnaire linkage that adds nothing. The lesson for educational data mining is methodological: distinguish a *feature* ceiling (movable, and diagnosable by what a richer feature family buys) from a *data* ceiling (a property of the measurement instrument), and use convergent evidence rather than a single learning curve to tell them apart. The distinction travels: the same feature ceiling appears in all nine systems and the amount each can move it scales with its between-school inequality ($r \approx 0.80$ with the ICC), and the data ceiling survives the obvious escape route, adding PISA's within-test process data lifts AUC to $\sim$0.86 but almost entirely through a cognitive-process signal endogenous to the score, leaving the deployable ceiling at $\sim$0.78. A plateau, then, can be moved by a new *feature family* or by a new *modality*, but only by data that a school could actually hold before the test.

Shared drivers, distinctive breadth. Albania's risk mechanism is not exotic. Math anxiety and school composition drive risk across the nine countries; what distinguishes Albania is that low proficiency is *broad*, sitting on a flat SES gradient and a low floor. This reframes the policy conversation away from targeting the poor (whom a flat gradient says are not disproportionately the whole problem) toward system-wide levers.

On causality, and the screener. Every model here is associational and fit on a single cross-section. The clearest illustration is anxiety: lower reported math anxiety co-occurs with higher scores, but it is partly a *consequence* of doing well, so a counterfactual that "reduces anxiety" in the model is not a defensible intervention estimate. We therefore ship the risk screener, calibrated probabilities, per-student SHAP drivers, and the fairness audit, as a *triage and communication* aid, not a lever simulator. The fairness result reinforces the point: a naive threshold over-flags the poorest quintile, so the tool is only defensible as a way to widen support; where equal false-positive treatment across SES is required, group-specific thresholds close the gap from 0.63 to 0.003 at a measured recall cost, an equity choice we surface rather than bury. The one causal question we could pose from observational data, whether the 2019 earthquake localised the collapse, we tested and answered in the negative: a difference-in-differences whose placebo fails identifies the loss as national, a well-identified null that guards against reading a convenient single-region story into a country-wide shock.

Limitations. The student questionnaire is the only instrument here; we have no teacher or parent files, and the school questionnaire, while linked, adds no signal beyond composition. Albania lacks ESCS in 2012 and 2015, limiting its own cross-cycle socioeconomic comparison (the peer-benchmarked and 2018-vs-2022 analyses are unaffected). Design-based standard errors are available only for the 2015–2022 cycles; the fixed-width 2009/2012 cycles fall back to an imputation-only SE. The forecast rests on five cycles and one structural break, which is why we report scenarios rather than a number. Finally, the $\sim$0.78 ceiling is specific to background-questionnaire prediction of a binary Level-2 target; we tested one richer modality, PISA's within-test computer-based process data, and found it moves the plateau only through a signal endogenous to the score, so genuinely exogenous process data that a school holds before the assessment (attendance, formative assessment, longitudinal records) remains the untested route and the natural next step. The difference-in-differences likewise licenses only a negative causal claim; a design that could license a positive policy lever still requires data we do not have.

8. Conclusion

Three in four Albanian fifteen-year-olds were below basic mathematics proficiency in 2022, and the number with which we began now has an explanation. The collapse was not a uniformly weaker cohort but a change in the structure of who is at risk and why: a domain classifier separates the 2018 and 2022 cohorts almost perfectly from background data alone, the variable that moved most was the support students receive in school rather than the resources they have at home, and the decline sits on the flattest socioeconomic gradient of the nine countries we compare. Read against Albania's history, a destructive 2019 earthquake, an extended pandemic closure, a thin remote-learning infrastructure, and the steady emigration of teachers and working-age adults, the statistical shape and the economic record tell the same story: a supply-side disruption to schooling, not a demand-side collapse in family means.

For anyone who would predict this risk, the practical lesson is where the ceiling comes from. Individual background answers plateau at about 0.73 AUC and no amount of tuning moves them; the single missing ingredient is the socioeconomic composition of a student's school, which lifts every model to about 0.78, after which a multilevel model and a direct school-questionnaire linkage independently confirm that the ceiling is now a property of what background questionnaires can measure rather than of the model. A plateau, in other words, can be a feature ceiling or a data ceiling, and telling them apart takes convergent evidence rather than one more round of optimisation.

For anyone who would act on it, the lesson is that a screener trained on a single cross-section reports associations, not levers. The same analysis that ranks math anxiety highly also shows why reducing a model input is not an intervention estimate, and the fairness audit shows a naive threshold over-flagging the poorest students. The tool we release is therefore built to widen support, never to ration it. The policy direction that follows is not targeted transfers to the poor, whom a flat gradient says are not the whole problem, but the rebuilding of schooling capacity for everyone: retaining teachers, reconstructing schools, and funding catch-up instruction.

Two questions that earlier framings left open we were able to take a first pass at here, and both answers point the same way. Richer data *can* move the 0.78 ceiling, PISA's within-test process data reaches ∼0.86, but only through a signal endogenous to the score itself, so the deployable ceiling holds and the genuinely exogenous process records (attendance, formative assessment, longitudinal files) that would capture the disruption directly remain the route worth building. And the one causal question the observational data admit, whether the 2019 earthquake localised the collapse, returns a well-identified null: a difference-in-differences with a failing placebo shows the loss was national, not concentrated at the epicentre. What remains is a design that can license a *positive* policy lever rather than rule a single-region story out. Albania's 2022 result is the clearest natural signal yet of what a compounded shock does to a fragile education system; understanding it well is the difference between planning for recovery and mistaking a broken process for a weak generation of students.

Reproducibility. Code, configuration, the unit-tested pipeline, the calibrated bilingual screener, and this manuscript's figures are in the project repository. Raw PISA microdata are public from the OECD but not redistributed here.

References

[1] Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proc. 22nd ACM SIGKDD Int. Conf. on Knowledge Discovery and Data Mining*, pages 785–794, 2016. doi: 10.1145/2939672.2939785.

[2] Elizabeth R. DeLong, David M. DeLong, and Daniel L. Clarke-Pearson. Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach. *Biometrics*, 44(3):837–845, 1988. doi: 10.2307/2531595.

[3] Marcos Delprato. Identifying the post-pandemic determinants of low performing students in latin america through interpretable machine learning methods. arXiv preprint, 2025. arXiv:2509.24508.

[4] Denis Djekourmane, Wei Zhang, Millicent Aziku, Dongfang Wu, and Yang Zhang. Using machine learning to predict student mathematics performance in six east asian countries: Evidence from PISA 2022. *Frontiers in Psychology*, 17, 2026. doi: 10.3389/fpsyg.2026.1840049.

[5] Seçil Erdoğan and Hüseyin Taştan. Predicting student achievement via machine learning: Evidence from the turkish subset of PISA. *Yildiz Social Science Review*, 10(1):7–27, 2024. doi: 10.51803/yssr.1461030.

[6] Harvey Goldstein. *Multilevel Statistical Models.* John Wiley & Sons, Chichester, 4th edition, 2011.

[7] Government of Albania and European Union and United Nations and World Bank. Albania post-disaster needs assessment: Volume a, following the 26 november 2019 earthquake. Technical report, Government of Albania, Tirana, 2020.

[8] Moritz Hardt, Eric Price, and Nathan Srebro. Equality of opportunity in supervised learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 29, pages 3315–3323, 2016.

[9] INSTAT (Institute of Statistics of Albania). Albanian population and housing census 2023. INSTAT, Tirana, 2024. `https://www.instat.gov.al/en/publications/books/2024/albanian-population-and-housing-census-2023/`.

[10] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, pages 3146–3154, 2017.

[11] Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, pages 4765–4774, 2017.

[12] Chiara Masci, Geraint Johnes, and Tommaso Agasisti. Student and school performance across countries: A machine learning approach. *European Journal of Operational Research*, 269(3):1072–1085, 2018. doi: 10.1016/j.ejor.2018.02.031.

[13] Claude Nadeau and Yoshua Bengio. Inference for the generalization error. *Machine Learning*, 52(3):239–281, 2003. doi: 10.1023/A:1024068626366.

[14] OECD. *PISA Data Analysis Manual: SPSS and SAS*. OECD Publishing, Paris, 2nd edition, 2009. doi: 10.1787/9789264056275-en.

[15] OECD. PISA 2022 Results (Volume I): The State of Learning and Equity in Education. Technical report, OECD Publishing, Paris, 2023.

[16] OECD. PISA 2022 Results (Volume I and II), Country Note: Albania. OECD Publishing, Paris, 2023. `https://www.oecd.org/en/publications/pisa-2022-results-volume-i-and-ii-country-notes_ed6fbcc5-en/albania_1ccc35b9-en.html`.

[17] Liudmila Prokhorenkova, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. CatBoost: Unbiased boosting with categorical features. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, pages 6638–6648, 2018.

[18] Joaquin Quiñonero-Candela, Masashi Sugiyama, Anton Schwaighofer, and Neil D. Lawrence. *Dataset Shift in Machine Learning*. MIT Press, Cambridge, MA, 2009.

[19] Donald B. Rubin. *Multiple Imputation for Nonresponse in Surveys*. John Wiley & Sons, New York, 1987.

[20] UNESCO. Global monitoring of school closures caused by the covid-19 pandemic. UNESCO Institute for Statistics, 2022. `https://covid19.uis.unesco.org/global-monitoring-school-closures-covid19/`.

[21] UNICEF Albania. The impact of covid-19 on children's learning in albania. Technical report, UNICEF, Tirana, 2021.

[22] Andrew J. Vickers and Elena B. Elkin. Decision curve analysis: A novel method for evaluating prediction models. *Medical Decision Making*, 26(6):565–574, 2006. doi: 10.1177/0272989X06295361.

[23] World Bank. Gdp growth (annual %), albania. world development indicators. World Bank Open Data, 2024. `https://data.worldbank.org/indicator/NY.GDP.MKTP.KD.ZG?locations=AL`.

[24] World Bank. Personal remittances, received (% of gdp), albania. world development indicators. World Bank Open Data, 2024. `https://data.worldbank.org/indicator/BX.TRF.PWKR.DT.GD.ZS?locations=AL`.

[25] World Bank and UNESCO and UNICEF and USAID and FCDO and Bill & Melinda Gates Foundation. The state of global learning poverty: 2022 update. Technical report, World Bank, Washington, DC, 2022.

[26] World Bank and UNESCO Institute for Statistics. Government expenditure on education, total (% of gdp), albania. world development indicators. World Bank Open Data, 2024. `https://data.worldbank.org/indicator/SE.XPD.TOTL.GD.ZS?locations=AL`.

[27] Bianca Zadrozny and Charles Elkan. Transforming classifier scores into accurate multiclass probability estimates. In *Proc. 8th ACM SIGKDD Int. Conf. on Knowledge Discovery and Data Mining*, pages 694–699, 2002. doi: 10.1145/775047.775151.